

# The Nitrogen Group

This group includes different types of natural element – non-metals, semi-metals, and dense metals – as well as moscovium (Mc), an artificial element. The group is also known as “pnictogens”. This derives from the Greek word *pnígein*, which means “to choke” and refers to the potential toxicity of nitrogen (N) in certain forms.



## Atomic structure

Members of this group have atoms with five electrons in the outer shell. These atoms can form up to three bonds at the same time.



## Physical properties

All the members are solids, except nitrogen (N). The density of elements increases down the group: bismuth (Bi) is 8,000 times denser than nitrogen.



## Chemical properties

Phosphorus (P), which exists in two main forms, is reactive, but the others in this group are fairly stable.



## Compounds

When reacting with three hydrogen (H) atoms, all members of the group form reactive gaseous compounds called hydrides.